

Generation Interconnection System Impact Study Report

For Queue Project Q17-0793

Hebron Rosemeade 100 kV

June 28, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0793. The Interconnection Customer proposes a 74.3 MW solar photovoltaic facility interconnecting at the Hebron Rosemeade 100 kV substation in the Indiana area. The request was received on May 07, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 231 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,536,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
HILLCREST – MULDOON MILL 100.0 kV Line	WAYNE TRACE – DARLINGTON RD 100.0 kV	132.6	210	7.8
MCNEIL SUBSTATION – BREED 230.0 kV	RICHLAND – DEKALB 500.0 kV	131.6	1,103	12.5
SIDNEY A MURRAY JR HYDROELECTRIC – BREED 230.0 kV	RICHLAND – DEKALB 500.0 kV	131.6	1,103	12.5
TWIN COUNTY RANCH – RICHLAND 230.0 kV L	TROY ENERGY LLC – RICHLAND 500.0 kV	125.9	822	6.3
MINNEHAHA – SIDNEY A MURRAY JR HYDROELECTRIC 230.0 kV Line	DEKALB – RICHLAND 500.0 kV	117.3	844	7.9
TROY ENERGY LLC SUBSTATION – HIGHGATE 230.0 kV line	TREGO – TROY ENERGY LLC 500.0 kV	116.4	755	5.7
CORRIDOR – LARSON ST 230.0 kV	TREGO – BEAVER 500.0 kV	115.6	640	8.8
E METHUEN 500/230 kV Transformer Bank	TREGO – BEAVER 500.0 kV	114.8	1,605	17.7
DARLINGTON ROAD – MUSKOGEE 100.0 kV	RICHLAND – DEKALB 500.0 kV	114.0	326	8.5
NAPLEON PEAKING STATION – RICHLAND 230.0 kV	RICHAND – TROY ENERGY LLC 500.0 kV	113.9	702	5.6
WAYNE TRACE – ALLEN 100.0 kV	SEGS 5 – HILLCREST 100.0 kV	113.9	298	8.0
TRUMAN – WALDO 230.0 kV	BEAVER – TREGO 500.0 kV Line	113.8	726	8.8
SEGS 5 – AEP PETTIT AVENUE 100.0 kV	ALLEN – WAYNE TRACE 100.0 kV	112.1	138	11.4
EUFAULA 230-1 kV Autotransformer	IMPERIAL – EUFAULA 500.0 kV line	111.3	1,680	11.8

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
EUFAULA 500 kV/1 kV Transformer	IMPERIAL – EUFAULA 500.0 kV	111.3	1,680	11.8
ADIRONDACK – JAMES DE YOUNG 230.0 kV line	ST. LAWRENCE – JAMES DE YOUNG 230.0 kV	110.7	748	8.4
DEAN FOREST – EASTOWN ROAD 230.0 kV	SWAN VLY – MUSKOGEE 500.0 kV line	110.5	643	9.0
MUSKOGEE – DEAN FOREST 230.0 kV	SWAN VLY – MUSKOGEE 500.0 kV	109.9	659	9.0
WAYNE TRACE – DARLINGTON D 100.0 kV Line	RICHLAND SUBSTATION – DEKALB 500.0 kV	108.2	371	7.9
BARAGA VLG – MADELINE 230.0 kV	GRAY’S FLAT – EAST METHUEN 230.0 kV L	107.8	514	7.7
STA S VAN NUYS – MINNEHAHA 230.0 kV	TROY ENERGY LLC – RICHLAND 500.0 kV	107.5	1,081	9.1
TREGO – TROY ENERGY LLC 500.0 kV	SWAN VLY – TREGO 500.0 kV	107.1	2,800	21.5
WALDO 500 kV/230 kV XFMR	TREGO – BEAVER SUBSTATION 500.0 kV Line	105.9	894	8.8
IMPERIAL – EUFAULA 500.0 kV	TREGO – GALION GENERATIG STA 500.0 kV line	105.4	2,771	26.8
JAMES DE YOUNG SUBSTATION – CARLEY RD 230.0 kV	EAST METHUEN SUBSTATION 500/230 kV Transformer Bank	105.0	647	8.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HILLCREST – MULDOON MILL 100.0 kV Line	Rebuild line with higher-capacity conductor	\$4,431,000	4% (\$191,000)
MCNEIL – BREED 230.0 kV line	Rebuild line with higher-capacity conductor	\$23,737,000	1% (\$177,000)
SIDNEY A MURRAY JR HYDROELECTRIC – BREED 230.0 kV	Rebuild line with higher-capacity conductor	\$51,920,000	1% (\$387,000)
TWIN COUNTY RANCH – RICHLAND 230.0 kV line	Rebuild line with higher-capacity conductor	\$36,868,000	0% (\$196,000)
MINNEHAHA – SIDNEY A MURRAY JR HYDROELECTRIC 230.0 kV Line	Rebuild line with higher-capacity conductor	\$30,792,000	1% (\$222,000)
TROY ENERGY LLC – HIGHGATE 230.0 kV	Rebuild line with higher-capacity conductor	\$53,263,000	1% (\$345,000)
CORRIDOR – LARSON ST 230.0 kV line	Rebuild line with higher-capacity conductor	\$20,650,000	1% (\$161,000)
EAST METHUEN 500-230 kV XFMR	Replace transformer with higher-rated unit	\$34,303,000	1% (\$246,000)
DARLINGTON ROAD – MUSKOGEE 100.0 kV	Rebuild line with higher-capacity conductor	\$12,179,000	2% (\$189,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
RICHLAND – NAPOLEON PEAKING STA 230.0 kV	Rebuild line with higher-capacity conductor	\$20,437,000	1% (\$124,000)
WAYNE TRACE – ALLEN 100.0 kV	Rebuild line with higher-capacity conductor	\$5,244,000	4% (\$184,000)
WALDO – TRUMAN 230.0 kV	Rebuild line with higher-capacity conductor	\$18,200,000	1% (\$142,000)
SEGS 5 – AEP PETTIT AVENUE 100.0 kV	Rebuild line with higher-capacity conductor	\$2,525,000	8% (\$209,000)
EUFAULA 230-1 kV Autotransformer	Replace transformer with higher-rated unit	\$11,064,000	1% (\$85,000)
EFAULA 500 kV/1 kV Auto	Replace transformer with higher-rated unit	\$38,179,000	1% (\$292,000)
ADIRONDACK – JAMES DE YOUNG 230.0 kV	Rebuild line with higher-capacity conductor	\$42,148,000	1% (\$313,000)
DEAN FOREST – EASTOWN ROAD 230.0 kV	Rebuild line with higher-capacity conductor	\$21,873,000	2% (\$401,000)
MUSKOGEE – DEAN FOREST 230.0 kV	Reconductor line	\$15,610,000	2% (\$286,000)
WAYNE TRACE – DARLINGTON D 100.0 kV Line	Reconductor line	\$671,000	2% (\$10,000)
BARAGA VLG – MADELINE 230.0 kV	Reconductor line	\$6,332,000	1% (\$43,000)
STA S VAN NUYS – MINNEHAHA 230.0 kV	Reconductor line	\$6,748,000	1% (\$49,000)
TREGO – TROY ENERGY LLC 500.0 kV line	Reconductor line	\$75,242,000	1% (\$536,000)
WALDO 500 kV/230 kV XFMR	Replace transformer with higher-rated unit	\$33,780,000	1% (\$262,000)
IMPERIAL – EUFAULA 500.0 kV	Reconductor line	\$53,991,000	1% (\$410,000)
CARLEY ROAD – JAMES DE YOUNG 230.0 kV	Reconductor line	\$10,999,000	1% (\$76,000)
	Total allocated to project		\$5,536,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.